

Utkarsh Singh Gunjial

Delhi, IN | utkarshsinghgunjial@gmail.com | 6239438842 | [GitHub](#) | [LinkedIn](#)

Education

Delhi Technological University, BTech in Computer Science
CGPA: 7.51/10

Sep 2022 – May 2026

Ishwar Singh Dev Samaj Sr. Sec. School, Chandigarh, PCM

March 2022

Experience

Business and Data Analytics Intern, TATA Motors Jamshedpur Plant

Sep 2025 – Nov 2025

- Built and deployed a Power BI dashboard with 12+ real-time visualizations monitoring key supply chain KPIs, reducing manual tracking time by 40%.
- Designed ETL pipelines over 100,000+ rows of raw logistics data with structured validation and transformation workflows to improve downstream analytics accuracy.

AI Engineer Intern, The Future University – Chandigarh, IN

May 2025 – Jul 2025

- Developed an institutional-grade AI stock scoring system evaluating 5000+ Indian equities using multi-pillar fundamental analysis and narrative risk sentiment; automated end-to-end via n8n agentic workflows and Gemini LLMs.
- Built 5+ agentic trading workflows with LLM-based multi-step reasoning for chart analysis, risk assessment, and trade evaluation; integrated Promptfoo for LLM evaluation and PostHog for observability, reducing manual steps by 30%.
- Deployed ChatGPT-powered Chart Analyser and Portfolio Analyser with structured prompt logic and iterative evaluation pipelines, serving 4,000+ active users.

Projects

Full-Stack AI Assistant Platform (React, FastAPI, MongoDB)

[GitHub](#) | [Live Demo](#)

- Designed and deployed a full-stack AI assistant using React, FastAPI, MongoDB, and JWT authentication supporting persistent multi-user conversations.
- Built multi-agent orchestration workflows combining retrieval, reasoning, evaluation, and response refinement to improve answer quality.
- Implemented document upload, OCR-enabled processing, FAISS vector search, hybrid retrieval, and thread-aware memory for context-aware interactions.
- Developed modular REST APIs for authentication, chat, document ingestion, retrieval, conversation history, and user session management.
- Integrated LangSmith tracing and observability for monitoring agent execution, debugging workflows, and analyzing system performance in production.

Human-in-the-Loop Invoice Intelligence Platform

[GitHub](#) | [Live Demo](#)

- Built and deployed a full-stack invoice processing platform using React, FastAPI, MongoDB Atlas, Render, and Vercel.
- Developed PDF/TXT ingestion and extraction pipelines with automated invoice validation, duplicate detection, and business rule enforcement.
- Implemented human-in-the-loop review workflows for high-value, low-confidence, and incomplete invoices with audit tracking.
- Designed JWT authentication, role-aware dashboards, invoice lifecycle management, REST APIs, CORS, and production environment configuration.

Smart Traffic Monitoring Multi-Object Tracking System

- Developed a computer vision pipeline using YOLOv8, OpenCV, and SORT tracking for real-time vehicle detection and monitoring.
- Implemented Kalman Filter-based object tracking to maintain stable vehicle identities across video streams.
- Built event detection and vehicle counting logic using trajectory-based line crossing algorithms.

Skills

- Languages:** Python, SQL, JavaScript, C++
- Frontend:** React, HTML, CSS
- Backend:** FastAPI, REST APIs, JWT Authentication, Async Programming
- Databases Storage:** MongoDB, SQL, FAISS, Vector Search
- AI Machine Learning:** PyTorch, Scikit-Learn, CNNs, NLP, Retrieval-Augmented Generation (RAG), Vector Embeddings, Agentic AI Workflows
- AI Frameworks Observability:** LangGraph, LangSmith, Promptfoo
- Cloud Deployment:** Vercel, Render, MongoDB Atlas, Git, GitHub
- Analytics Automation:** Power BI, Pandas, NumPy, ETL Pipelines, Data Validation, n8n